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AGGREGATION OF APPLICATION AND SYSTEMS FOR SERVICES FOR 6G OR OTHER NEXT GENERATION NETWORK

Abstract

In a 6G network, microservices can be utilized in the absence of a core network. For example, after a mobile device has authenticated, when the mobile device moves away from mobile edge computing (MEC) coverage, the information from the MEC can be ported to a new MEC along the way of the mobile device's path. The initial MEC can send service related application package data to one or more neighboring MECs. When a neighboring MEC of the neighboring MECs receives the service related application package data and the neighboring MEC serves the area which the mobile device is transitioning to, then the neighboring MEC can send an acknowledgment back to the initial MEC. Once the acknowledgment is received by the initial MEC, the initial MEC can complete the handover of the application data needed to reduce or mitigate any disruption in service for the mobile device.

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Background/Summary

RELATED APPLICATION [0001] The subject patent application is a continuation of, and claims priority to, U.S. patent application Ser. No. 17/805,118, filed Jun. 2, 2022, which is a continuation of, and claims priority to, U.S. patent application Ser. No. 16/568,884, filed Sep. 12, 2019, and entitled “AGGREGATION OF APPLICATION AND SYSTEMS FOR SERVICES FOR 6G OR OTHER NEXT GENERATION NETWORK,” the entirety of which priority applications are hereby incorporated by reference herein.

TECHNICAL FIELD

[0002] This disclosure relates generally to aggregation and multiplication of applications and services. For example, this disclosure relates to facilitating aggregated services via edge computing for a 6G, or other next generation network.

BACKGROUND

[0003] Microservices are a software development technique—a variant of the service-oriented architecture (SOA) architectural style that structures an application as a collection of loosely coupled services. In a microservices architecture, services are fine-grained and the protocols are lightweight. The benefit of decomposing an application into different smaller services is that it improves modularity. This makes the application easier to understand, develop, test, and become more resilient to architecture erosion. It parallelizes development by enabling small autonomous teams to develop, deploy and scale their respective services independently. It also allows the architecture of an individual service to emerge through continuous refactoring. Microservice-based architectures enable continuous delivery and deployment.

[0004] The above-described background relating facilitating aggregated services via edge computing is not intended to be exhaustive. Other contextual information may become further apparent upon review of the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Non-limiting and non-exhaustive embodiments of the subject disclosure are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified.

[0006] FIG. 1 illustrates an example wireless communication system in which a network node device (e.g., network node) and user equipment (UE) can implement various aspects and embodiments of the subject disclosure.

[0007] FIG. 2 illustrates an example schematic system block diagram of a mobile edge computing component according to one or more embodiments.

[0008] FIG. 3 illustrates an example schematic system block diagram of a mobile application function component according to one or more embodiments.

[0009] FIG. 4 illustrates an example aggregated services mobile edge computing system according to one or more embodiments.

[0010] FIG. 5 illustrates an example system flow diagram for aggregated services via mobile edge computing according to one or more embodiments.

[0011] FIG. 6 illustrates an example flow diagram of a method for facilitating aggregated services via edge computing according to one or more embodiments.

[0012] FIG. 7 illustrates an example flow diagram of a system for facilitating aggregated services via edge computing according to one or more embodiments.

[0013] FIG. 8 illustrates an example flow diagram of a machine-readable medium for facilitating aggregated services via edge computing according to one or more embodiments.

[0014] FIG. 9 illustrates an example block diagram of an example mobile handset operable to engage in a system architecture that facilitates secure wireless communication according to one or more embodiments described herein.

[0015] FIG. 10 illustrates an example block diagram of an example computer operable to engage in a system architecture that facilitates secure wireless communication according to one or more embodiments described herein.

DETAILED DESCRIPTION

[0016] In the following description, numerous specific details are set forth to provide a thorough understanding of various embodiments. One skilled in the relevant art will recognize, however, that the techniques described herein can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring certain aspects.

[0017] Reference throughout this specification to “one embodiment,” or “an embodiment,” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase “in one embodiment,” “in one aspect,” or “in an embodiment,” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0018] As utilized herein, terms “component,” “system,” “interface,” and the like are intended to refer to a computer-related entity, hardware, software (e.g., in execution), and/or firmware. For example, a component can be a processor, a process running on a processor, an object, an executable, a program, a storage device, and/or a computer. By way of illustration, an application running on a server and the server can be a component. One or more components can reside within a process, and a component can be localized on one computer and/or distributed between two or more computers.

[0019] Further, these components can execute from various machine-readable media having various data structures stored thereon. The components can communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network, e.g., the Internet, a local area network, a wide area network, etc. with other systems via the signal).

[0020] As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry; the electric or electronic circuitry can be operated by a software application or a firmware application executed by one or more processors; the one or more processors can be internal or external to the apparatus and can execute at least a part of the software or firmware application. As yet another example, a component can be

an apparatus that provides specific functionality through electronic components without mechanical parts; the electronic components can include one or more processors therein to execute software and/or firmware that confer(s), at least in part, the functionality of the electronic components. In an aspect, a component can emulate an electronic component via a virtual machine, e.g., within a cloud computing system.

[0021] The words “exemplary” and/or “demonstrative” are used herein to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as “exemplary” and/or “demonstrative” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques known to those of ordinary skill in the art. Furthermore, to the extent that the terms “includes,” “has,” “contains,” and other similar words are used in either the detailed description or the claims, such terms are intended to be inclusive—in a manner similar to the term “comprising” as an open transition word—without precluding any additional or other elements.

[0022] As used herein, the term “infer” or “inference” refers generally to the process of reasoning about, or inferring states of, the system, environment, user, and/or intent from a set of observations as captured via events and/or data. Captured data and events can include user data, device data, environment data, data from sensors, sensor data, application data, implicit data, explicit data, etc. Inference can be employed to identify a specific context or action, or can generate a probability distribution over states of interest based on a consideration of data and events, for example.

[0023] Inference can also refer to techniques employed for composing higher-level events from a set of events and/or data. Such inference results in the construction of new events or actions from a set of observed events and/or stored event data, whether the events are correlated in close temporal proximity, and whether the events and data come from one or several event and data sources.

Various classification schemes and/or systems (e.g., support vector machines, neural networks, expert systems, Bayesian belief networks, fuzzy logic, and data fusion engines) can be employed in connection with performing automatic and/or inferred action in connection with the disclosed subject matter.

[0024] In addition, the disclosed subject matter can be implemented as a method, apparatus, or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device, machine-readable device, computer-readable carrier, computer-readable media, or machine-readable media. For example, computer-readable media can include, but are not limited to, a magnetic storage device, e.g., hard disk; floppy disk; magnetic strip(s); an optical disk (e.g., compact disk (CD), a digital video disc (DVD), a Blu-ray Disc™ (BD)); a smart card; a flash memory device (e.g., card, stick, key drive); and/or a virtual device that emulates a storage device and/or any of the above computer-readable media.

[0025] As an overview, various embodiments are described herein to facilitate aggregated services via edge computing for a 6G interface or other next generation networks. For simplicity of explanation, the methods (or algorithms) are depicted and described as a series of acts. It is to be understood and appreciated that the various embodiments are not limited by the acts illustrated and/or by the order of acts. For example, acts can occur in various orders and/or concurrently, and with other acts not presented or described herein. Furthermore, not all illustrated acts may be required to implement the methods. In addition, the methods could alternatively be represented as a series of interrelated states via a state diagram or events. Additionally, the methods described hereafter are capable of being stored on an article of manufacture (e.g., a machine-readable storage medium) to facilitate transporting and transferring such methodologies to computers. The term article of manufacture, as used herein, is intended to encompass a computer program accessible

from any computer-readable device, carrier, or media, including a non-transitory machine-readable storage medium.

[0026] It should be noted that although various aspects and embodiments have been described herein in the context of 6G, the disclosed aspects are not limited to 6G, a UMTS implementation, and/or an LTE implementation as the techniques can also be applied in 3G, 4G, 5G, or LTE systems. For example, aspects or features of the disclosed embodiments can be exploited in substantially any wireless communication technology. Such wireless communication technologies can include UMTS, Code Division Multiple Access (CDMA), Wi-Fi, Worldwide Interoperability for Microwave Access (WiMAX), General Packet Radio Service (GPRS), Enhanced GPRS, Third Generation Partnership Project (3GPP), LTE, Third Generation Partnership Project 2 (3GPP2) Ultra Mobile Broadband (UMB), High Speed Packet Access (HSPA), Evolved High Speed Packet Access (HSPA+), High-Speed Downlink Packet Access (HSDPA), High-Speed Uplink Packet Access (HSUPA), Zigbee, or another IEEE 802.XX technology. Additionally, substantially all aspects disclosed herein can be exploited in legacy telecommunication technologies.

[0027] Described herein are systems, methods, articles of manufacture, and other embodiments or implementations that can facilitate aggregated services via edge computing for a 6G network. Facilitating aggregated services via edge computing for a 6G network can be implemented in connection with any type of device with a connection to the communications network (e.g., a mobile handset, a computer, a handheld device, etc.) any Internet of things (IOT) device (e.g., toaster, coffee maker, blinds, music players, speakers, etc.), and/or any connected vehicles (cars, airplanes, space rockets, and/or other at least partially automated vehicles (e.g., drones)). In some embodiments the non-limiting term user equipment (UE) is used. It can refer to any type of wireless device that communicates with a radio network node in a cellular or mobile communication system. Examples of UE are target device, device to device (D2D) UE, machine type UE or UE capable of machine to machine (M2M) communication, PDA, Tablet, mobile terminals, smart phone, laptop embedded equipped (LEE), laptop mounted equipment (LME), USB dongles etc. Note that the terms element, elements and antenna ports can be interchangeably used but carry the same meaning in this disclosure. The embodiments are applicable to single carrier as well as to multicarrier (MC) or carrier aggregation (CA) operation of the UE. The term carrier aggregation (CA) is also called (e.g., interchangeably called) “multi-carrier system”, “multi-cell operation”, “multi-carrier operation”, “multi-carrier” transmission and/or reception.

[0028] In some embodiments the non-limiting term radio network node or simply network node is used. It can refer to any type of network node that serves UE is connected to other network nodes or network elements or any radio node from where UE receives a signal. Examples of radio network nodes are Node B, base station (BS), multi-standard radio (MSR) node such as MSR BS, eNode B, network controller, radio network controller (RNC), base station controller (BSC), relay, donor node controlling relay, base transceiver station (BTS), access point (AP), transmission points, transmission nodes, RRU, RRH, nodes in distributed antenna system (DAS) etc.

[0029] Cloud radio access networks (RAN) can enable the implementation of concepts such as software-defined network (SDN) and network function virtualization (NFV) in 6G networks. This disclosure can facilitate a generic channel state information framework design for a 6G network. Certain embodiments of this disclosure can comprise an SDN controller that can control routing of traffic within the network and between the network and traffic destinations. The SDN controller can be merged with the 6G network architecture to enable service deliveries via open application programming interfaces (“APIs”) and move the network core towards an all internet protocol (“IP”), cloud based, and software driven telecommunications network. The SDN controller can work with, or take the place of policy and charging rules function (“PCRF”) network elements so that policies such as quality of service (QoS) and traffic management and routing can be synchronized and managed end to end.

[0030] An LTE network can be a policy-based traffic management architecture with a PCRF

element traditionally controlling the QoS levels and other information (priorities bandwidths, etc.) that manages IP flows that carries a particular application (such as voice, video, messaging, etc.). This policy-based mechanism applies to the IP traffic between the mobile device and the packet data network gateway ("PGW"). In an embodiment of the subject disclosure, software defined networking can be used to provide routing and traffic control for packets sent from the PGW to a destination address. In some embodiments, the SDN controller can also provide traffic control for packets from the mobile device to the destination in some embodiments.

[0031] The PCRF and the SDN controller can also communicate about some aspects of a particular application flow so that routing decisions both in the access network (between eNodeB and PGW) as well as in the backbone can be made based on the nature of the application and how that particular flow was expected to be treated based on operator policies and user subscription. For example, if a higher QoS is to be applied to a traffic flow carrying voice packet, the service related information such as QoS can be used by SDN controller to make decisions such as mapping and route optimizations. This can enable the entire network to be application aware with a consistent treatment of the packets.

[0032] Radio access network abstraction can provide a separation between the physical radios and a logical view of the network. It can provide a holistic view of a pool of various radio resources from various radio technologies. This can allow a network controller to make an intelligent decision on what radio to use to deliver a service based on application requirements. The radio access network abstraction can also have a dynamic learning capability to constantly update the network view of the radio resources upon adding, changing, removing and/or modifying the resources.

[0033] A 6G network has the ability to dedicate an edge slice with the capability to intelligently perform edge computing of a large number of information on demand. In a 6G network, microservice enabled solutions can bypasses the core network. Additionally, after an initial provisioning the network can autonomously communicate with connected parties. This solution can utilize a dynamic handling request for packet propagation. Dynamic means that the handling request can change depending on the distance the packet has to travel, time associated with travel, time of day, time of year, etc.

[0034] For example, if a packet is set to travel via the network and interact with several nodes, the sending node can create a profile for the packet and attach a packet profile to the packet according to packet characteristics. The packet profile can comprise initial packet profile characteristics and be distributed within a 6G edge slice tailored for a specific service associated with the packet. Once the packet is ready to be sent, the packet profile can be sent, with and/or in advance of the packet itself, to downstream nodes. Consequently, each node can be informed and updated with every reading of the packet and the possibility of a false positive with regards to the packet characteristics can be mitigated or eliminated.

[0035] In a 6G network, microservices can be utilized as an alternative to the core network. For example, after a device is on-boarded (e.g., path set-up, authentication, level of service, etc.) to the network, the microservices can facilitate internetwork communication. Thus, microservices can perform certain functions without the core network (e.g., changing prices, latency mitigation, etc.).

[0036] In a modem access network with numerous access technology connecting billions of devices to the core agnostic network, there are certain intelligence that can be used at the access to accommodate the services to communicate with subscriber UE devices. These UE devices can be simple as a connected light bulb or a complicated and integrated connected car on board unit (OBU) with sub processors integrated to various integral parts of a vehicle. Some of these services running on the UEs can either be tailored to specific applications running on the edge of the network or information that need to enable services are available at the edge or at the mobile edge computing (MEC). To accommodate these services, a MEC aggregation capability can be utilized. MEC aggregation can be performed when a device connected to a MEC moves away from MEC

coverage. The information in the MEC and what applications need to be instantiated in the MEC can be ported to new MEC along the pathway of the UE device as it moves.

[0037] Currently access networks are evolving with more radio technologies on high throughput technology such as millimeter wave (mmW) as well as technologies such as long-term evolution mobile (LteM). Services are also evolving to consume more data and offer a plethora of streaming possibilities such as demanding virtual reality streaming and high end interactive augmented reality solutions. A large number of these services can already depend on micro services running at the edge of the network, or in a near future the services can share the capacity of different access technologies simultaneously. In this disclosure, the network can build around such applications based on the MEC closest to the UE device. Once a service is requested, the application can get instantiated with necessary co apps and databases. However, if the user is moving out of the MEC coverage, the process can either be repeated or the service rely on the limited apps installed in the UE to partly substitute for the lack of edge application for that specific service. This can result in less than satisfactory experience and, in most cases, termination of the enhanced service that usually needs the extra information from the network for a tailboard experience, commercial ads, and/or other metadata that goes with the service.

[0038] As the access network evolves to the next generation of networks, there is a desire to implement an edge network to accommodate moving data from high demanding services between different access networks without degradation and interruption of moving UE between different radio technologies and coverage of radios. As a use case, when a subscriber initiates a service, a number of applications can get instantiated in the edge computing unit to accompany and enhance the given service. This service can be a simple service such as, a mapping service that guides a driver to a destination (e.g., concert, theater, movies, etc.). The accompanied application can be a database with tailored data about the subscriber's (e.g., drivers) likes and preferences as well as information about their destination (e.g., favorite restaurants) or even information about the play they are going to see. While the subscriber starts driving to his destination, the hosting MEC can have all the applications and databases needed for that specific service. The MEC host micro-services (e.g., applications necessary for the specific service hosted from the carrier or 3rd party service provider), as part of a radio access network intelligent controller (RIC) having a database that is partly instantiated from the user profile and partly populated in real-time by gathering information from the subscribers devices and surrounding sensor devices. This can provide the subscriber with a rich experience and provide a service provider an opportunity to gather information or offer add-on services. To be able to give both the user and the service provider a continuity and quality of experience (QoE) for during the trip, all MEC device subscribers passing through can utilize this package of applications and dynamic database. To accommodate these services, the MEC aggregation capability be utilized. The MEC aggregation can occur when a device connected to a MEC moves away from MEC coverage. Consequently, the information in the MEC, and what applications need to be instantiated in MEC, can be packaged and ported to a new MEC along the route of the UE device.

[0039] In one embodiment, described herein is a method comprising receiving, by a first network device of a wireless network and comprising a processor, location data representative of a location of a mobile device. The method can comprise generating, by the first network device, service application package data representative of a service the mobile device is utilizing in response to the receiving the location data. Additionally, in response to the generating the service application package data, the method can comprise sending, by the first network device, the service application package data to a second network device of the wireless network.

[0040] According to another embodiment, a system can facilitate obtaining location data representative of a location of a mobile device. In response to the obtaining the location data, the system can comprise generating service application package data representative of a service the mobile device has been determined to have utilized. Furthermore, in response to the generating the

service application package data, the system can comprise facilitating a transfer of a wireless connection of the mobile device from being connected to the first network device to being connected to a second network device of the wireless network.

[0041] According to yet another embodiment, described herein is a machine-readable storage medium that can perform the operations comprising receiving location data representative of a location of a mobile device. The machine-readable storage medium can perform the operations comprising receiving first service application data representative of a first service the mobile device has been determined to have utilized from the mobile device in response to the receiving the location data. Furthermore, in response to the receiving the first service application data, the machine-readable storage medium can perform the operations comprising aggregating the first service application data with second service application data, resulting in aggregated service application data. Additionally, in response to the aggregating, the machine-readable storage medium can perform the operations comprising facilitating a handover of the mobile device from the first network device to a second network device of the wireless network.

[0042] These and other embodiments or implementations are described in more detail below with reference to the drawings.

[0043] Referring now to FIG. 1, illustrated is an example wireless communication system **100** in accordance with various aspects and embodiments of the subject disclosure. In one or more embodiments, system **100** can comprise one or more user equipment UEs **102**. The non-limiting term user equipment can refer to any type of device that can communicate with a network node in a cellular or mobile communication system. A UE can have one or more antenna panels having vertical and horizontal elements. Examples of a UE comprise a target device, device to device (D2D) UE, machine type UE or UE capable of machine to machine (M2M) communications, personal digital assistant (PDA), tablet, mobile terminals, smart phone, laptop mounted equipment (LME), universal serial bus (USB) dongles enabled for mobile communications, a computer having mobile capabilities, a mobile device such as cellular phone, a laptop having laptop embedded equipment (LEE, such as a mobile broadband adapter), a tablet computer having a mobile broadband adapter, a wearable device, a virtual reality (VR) device, a heads-up display (HUD) device, a smart car, a machine-type communication (MTC) device, and the like. User equipment UE **102** can also comprise IOT devices that communicate wirelessly. It should be noted that the UE **102** can be a mobile device **102**.

[0044] In various embodiments, system **100** is or comprises a wireless communication network serviced by one or more wireless communication network providers. In example embodiments, a UE **102** can be communicatively coupled to the wireless communication network via a network node **104**. The network node (e.g., network node device) can communicate with user equipment (UE), thus providing connectivity between the UE and the wider cellular network. The UE **102** can send transmission type recommendation data to the network node **104**. The transmission type recommendation data can comprise a recommendation to transmit data via a closed loop MIMO mode and/or a rank-1 precoder mode.

[0045] A network node can have a cabinet and other protected enclosures, an antenna mast, and multiple antennas for performing various transmission operations (e.g., MIMO operations). Network nodes can serve several cells, also called sectors, depending on the configuration and type of antenna. In example embodiments, the UE **102** can send and/or receive communication data via a wireless link to the network node **104**. The dashed arrow lines from the network node **104** to the UE **102** represent downlink (DL) communications and the solid arrow lines from the UE **102** to the network nodes **104** represents an uplink (UL) communication.

[0046] System **100** can further include one or more communication service provider networks **106** that facilitate providing wireless communication services to various UEs, including UE **102**, via the network node **104** and/or various additional network devices (not shown) included in the one or more communication service provider networks **106**. The one or more communication service

provider networks **106** can include various types of disparate networks, including but not limited to: cellular networks, femto networks, picocell networks, microcell networks, internet protocol (IP) networks, Wi-Fi service networks, broadband service network, enterprise networks, cloud based networks, and the like. For example, in at least one implementation, system **100** can be or include a large scale wireless communication network that spans various geographic areas. According to this implementation, the one or more communication service provider networks **106** can be or include the wireless communication network and/or various additional devices and components of the wireless communication network (e.g., additional network devices and cell, additional UEs, network server devices, etc.). The network node **104** can be connected to the one or more communication service provider networks **106** via one or more backhaul links **108**. For example, the one or more backhaul links **108** can comprise wired link components, such as a T1/E1 phone line, a digital subscriber line (DSL) (e.g., either synchronous or asynchronous), an asymmetric DSL (ADSL), an optical fiber backbone, a coaxial cable, and the like. The one or more backhaul links **108** can also include wireless link components, such as but not limited to, line-of-sight (LOS) or non-LOS links which can include terrestrial air-interfaces or deep space links (e.g., satellite communication links for navigation).

[0047] Wireless communication system **100** can employ various cellular systems, technologies, and modulation modes to facilitate wireless radio communications between devices (e.g., the UE **102** and the network node **104**). While example embodiments might be described for 6G new radio (NR) systems, the embodiments can be applicable to any radio access technology (RAT) or multi-RAT system where the UE operates using multiple carriers e.g., LTE FDD/TDD, GSM/GERAN, CDMA2000, etc.

[0048] For example, system **100** can operate in accordance with global system for mobile communications (GSM), universal mobile telecommunications service (UMTS), long term evolution (LTE), LTE frequency division duplexing (LTE FDD, LTE time division duplexing (TDD), high speed packet access (HSPA), code division multiple access (CDMA), wideband CDMA (WCDMA), CDMA2000, time division multiple access (TDMA), frequency division multiple access (FDMA), multi-carrier code division multiple access (MC-CDMA), single-carrier code division multiple access (SC-CDMA), single-carrier FDMA (SC-FDMA), orthogonal frequency division multiplexing (OFDM), discrete Fourier transform spread OFDM (DFT-spread OFDM) single carrier FDMA (SC-FDMA), filter bank based multi-carrier (FBMC), zero tail DFT-spread-OFDM (ZT DFT-s-OFDM), generalized frequency division multiplexing (GFDM), fixed mobile convergence (FMC), universal fixed mobile convergence (UFMC), unique word OFDM (UW-OFDM), unique word DFT-spread OFDM (UW DFT-Spread-OFDM), cyclic prefix OFDM CP-OFDM, resource-block-filtered OFDM, Wi Fi, WLAN, WiMax, and the like. However, various features and functionalities of system **100** are particularly described wherein the devices (e.g., the UEs **102** and the network node **104**) of system **100** are configured to communicate wireless signals using one or more multi carrier modulation schemes, wherein data symbols can be transmitted simultaneously over multiple frequency subcarriers (e.g., OFDM, CP-OFDM, DFT-spread OFDM, UFMC, FBMC, etc.). The embodiments are applicable to single carrier as well as to multicarrier (MC) or carrier aggregation (CA) operation of the UE. The term carrier aggregation (CA) is also called (e.g., interchangeably called) “multi-carrier system”, “multi-cell operation”, “multi-carrier operation”, “multi-carrier” transmission and/or reception. Note that some embodiments are also applicable for Multi RAB (radio bearers) on some carriers (that is data plus speech is simultaneously scheduled).

[0049] Referring now to FIG. 2, illustrated is an example schematic system block diagram of a mobile edge computing component **200**. The mobile edge computing (MEC) component **200** can comprise a content component **202**, a user plane function component **204**, a session manager function component **206**, an updating component **208**, a microservices component **210**, a database **212**, memory **216**, and a processor **214**, which can all be communicatively coupled. The processor

214 can correspond to a processing component from a plurality of processing components. Aspects of the processor **214** can constitute machine-executable component(s) embodied within machine(s), e.g., embodied in one or more computer readable mediums (or media) associated with one or more machines. Such component(s), when executed by the one or more machines, e.g., computer(s), computing device(s), virtual machine(s), etc. can cause the machine(s) to perform the operations described. In an aspect, the processor **214** can also include memory **216** that stores computer executable components and instructions.

[0050] The service network can add some of its elements into the edge network depending on what services (e.g., microservices) are being used and/or called upon. A microservices component **210** can add additional capabilities to ongoing services between the service layer and subscriber devices (e.g., UE **102**). For example, the microservices component **210** can support content that is generated by the content component **202**. However, when a UE **102** is using certain microservices, there is certain data that can be exchanged so that the database **212** can be utilized to store information about these microservices. The user plane function component **204** comprises the data that the UE **102** is consuming, which can change at any time. However, this information can be stored at the database **212** via a caching function. Thus, the data can be stored for a short period of time and utilized by any other services and updated by the updating component **208** over time as the services change. The session manager function (SMF) component **206** can set up a session between the core network and the UE **102** device. Thus, the session can utilize any of these technologies to accommodate the user's needs in an on-demand fashion.

[0051] Referring now to FIG. **3**, illustrated is an example schematic system block diagram of a mobile application function component **300**. The mobile application function component **300** can comprise a filter function component **302**, a policy function component **304**, an application function component **306**, a database **308**, a processor **310** and a memory **312**, which can all be communicatively coupled. The processor **310** can correspond to a processing component from a plurality of processing components. Aspects of the processor **310** can constitute machine-executable component(s) embodied within machine(s), e.g., embodied in one or more computer readable mediums (or media) associated with one or more machines. Such component(s), when executed by the one or more machines, e.g., computer(s), computing device(s), virtual machine(s), etc. can cause the machine(s) to perform the operations described. In an aspect, the processor **310** can also include memory **312** that stores computer executable components and instructions.

[0052] The mobile application function (MAFs) component **300** can comprise additional data that is specifically related to the subscriber (e.g., UE **102**) than the MEC component **200** can have. For example, the MAF component **300** can handle very detailed data exchanges between the UE **102** and the MEC component **200**. The MAF component **300** can comprise functionality allowing related applications to be ported and/or cascaded to other MAF components following the subscriber's physical movements. The MAF component **300** can also comprise a filtering function component **302** that filters the related data to the core service provider depending on policies and service level agreements (SLA). For example, some mobile application data may need to be sent to the service provider for authentication and/or other purposes, and other data can be filtered out to be exchanged via multiple MAF components. Furthermore, the MAF component **300** can comprise a policy function (PF) component **304** that comprises information that can be updated with the SLA (e.g., such as dynamic changes to the policy via the user or carrier can be transferred/updated in PF component) and other variables such as user-defined data, and carrier core policy data. Thus, the filter function component **302** can use the information from the policy function component **304** to filter data to and from the service providers. Thus, the MAF component **300** can tailor its information specifically to the subscriber (e.g., what services the subscriber is using, where the subscriber is headed, what tickets were purchased, points-of-interests, predicted points-of-interest, historical data, etc.)

[0053] Referring now to FIG. **4**, illustrated is an example aggregated services mobile edge

computing system **400**. The aggregated services mobile edge computing system **400** can comprise the MEC component **200** and MAF components **300A**, **300B**, **300C**. It should be noted that any number of MEC components **200** and/or MAF components **300** are possible. FIG. 5 should not be considered as limiting as it is only exemplary. Microservices can be aggregated and ported from one MAF to another MAF. For example, if a UE **102** is at a static location and streaming video data on the UE **102**, the subscriber can be utilizing a number of services (e.g., streaming services, shopping services, html services, etc.). However, if the subscriber gets ready to leave the static location and gets in a vehicle, the UE **102** can still be on the edge network. For example, the MEC component **204** and the MAF components **300A**, **300B**, **300C** can be assigned specific areas (e.g., geographic locations) and comprise multiple functions as mentioned above. Once the UE **102** begins to move, and gets out of the coverage of the MEC component **200**, then the information that is a part of the subscriber's service coverage can be leveraged at another MEC component (not shown) and/or another MAF component. For example, the microservices being utilized by the UE **102** can be ported to the MAF component **300C** as a service related application package (SRAP) and handed over to other MECs and/or other MAF components, which can allow the user to continue to utilize the microservices without any interruption. The SRAP can also be sent by the service provider to specific MAFs based on the coverage area associated with each MAF. For example, the changing of the longitude and/or latitude of a UE **102** can trigger the network to indicate to one MAF **300C** that the UE **102** is leaving its coverage area **402** and trigger another MAF component **300B** that the UE **102** is entering its coverage area **404** or another coverage area **406**. MAF components **300A**, **300B**, **300C** can also comprise a neighbor list of what other MAF components are around nearby. So, one MAF component **300B** can send a request to push data to another MAF component **300A**, **300C** (or multiple other MECs) and request an acknowledgment from the MAF component that has accepted the data transfer. Upon receipt of this acknowledgment, the MAF component **300B** can send the data (via the SRAP) about the subscriber to the MAF component **300A**, **300C** that has responded with the acknowledgment data.

[0054] Referring now to FIG. 5, illustrated is an example system flow diagram for aggregated services via mobile edge computing. At block **500**, the MEC component **200** can receive location data (e.g., global positioning system (GPS) coordinate data, previous location data, anticipated location data based on historical GPS data association with applications, etc.) from the UE **102**, and at block **502**, the MEC component **200** can receive application request data from the UE **102**. After receiving the application request data from the UE **102**, the MEC **200** can instantiate the application based on the application request data and the location data at block **504**. At decision block **506**, if the UE's location is changing or the MEC component **200** receives an indication that the UE's location is about to change, then the MEC component **200** can send the application data to a neighbor MAF component **300** at block **510** so that utilization of the application is not interrupted. However, if the UE's location is not changing or not about to change, then the MEC component **200** can continue supporting application usage by the UE **102**. In alternative embodiments, the various MAF components **300** can handoff SRAPs between each other to facilitate application usage.

[0055] Referring now to FIG. 6, illustrated is an example flow diagram of a method for facilitating aggregated services via edge computing according to one or more embodiments. At element **600**, a method can comprise receiving, by a first network device of a wireless network, location data representative of a location of a mobile device. At element **602**, the method can comprise generating, by the first network device, service application package data representative of a service the mobile device is utilizing in response to the receiving the location data. Additionally, at element **604**, in response to the generating the service application package data, the method can comprise sending, by the first network device, the service application package data to a second network device of the wireless network.

[0056] Referring now to FIG. 7, illustrated is an example flow diagram of a system for facilitating

aggregated services via edge computing according to one or more embodiments. At element **700**, a system can facilitate obtaining location data representative of a location of a mobile device. In response to the obtaining the location data, at element **702**, the system can comprise generating service application package data representative of a service the mobile device has been determined to have utilized. Furthermore, in response to the generating the service application package data, the system can comprise facilitating a transfer of a wireless connection of the mobile device from being connected to the first network device to being connected to a second network device of the wireless network at element **704**.

[0057] Referring now to FIG. **8**, illustrated is an example flow diagram of a machine-readable medium for facilitating aggregated services via edge computing according to one or more embodiments. At element **800**, a machine-readable storage medium that can perform the operations comprising receiving location data representative of a location of a mobile device. At element **802**, the machine-readable storage medium can perform the operations comprising receiving first service application data representative of a first service the mobile device has been determined to have utilized from the mobile device in response to the receiving the location data. Furthermore, at element **804**, in response to the receiving the first service application data, the machine-readable storage medium can perform the operations comprising aggregating the first service application data with second service application data, resulting in aggregated service application data at element. Additionally, at element **806**, in response to the aggregating, the machine-readable storage medium can perform the operations comprising facilitating a handover of the mobile device from the first network device to a second network device of the wireless network.

[0058] Referring now to FIG. **9**, illustrated is an example block diagram of an example mobile handset **900** operable to engage in a system architecture that facilitates wireless communications according to one or more embodiments described herein. Although a mobile handset is illustrated herein, it will be understood that other devices can be a mobile device, and that the mobile handset is merely illustrated to provide context for the embodiments of the various embodiments described herein. The following discussion is intended to provide a brief, general description of an example of a suitable environment in which the various embodiments can be implemented. While the description includes a general context of computer-executable instructions embodied on a machine-readable storage medium, those skilled in the art will recognize that this disclosure also can be implemented in combination with other program modules and/or as a combination of hardware and software.

[0059] Generally, applications (e.g., program modules) can include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods described herein can be practiced with other system configurations, including single-processor or multiprocessor systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0060] A computing device can typically include a variety of machine-readable media. Machine-readable media can be any available media that can be accessed by the computer and includes both volatile and non-volatile media, removable and non-removable media. By way of example and not limitation, computer-readable media can comprise computer storage media and communication media. Computer storage media can include volatile and/or non-volatile media, removable and/or non-removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, program modules, or other data. Computer storage media can include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD ROM, digital video disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the

computer.

[0061] Communication media typically embodies computer-readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of the any of the above should also be included within the scope of computer-readable media.

[0062] The handset includes a processor **902** for controlling and processing all onboard operations and functions. A memory **904** interfaces to the processor **902** for storage of data and one or more applications **906** (e.g., a video player software, user feedback component software, etc.). Other applications can include voice recognition of predetermined voice commands that facilitate initiation of the user feedback signals. The applications **906** can be stored in the memory **904** and/or in a firmware **908**, and executed by the processor **902** from either or both the memory **904** or/and the firmware **908**. The firmware **908** can also store startup code for execution in initializing the handset **900**. A communications component **910** interfaces to the processor **902** to facilitate wired/wireless communication with external systems, e.g., cellular networks, VoIP networks, and so on. Here, the communications component **910** can also include a suitable cellular transceiver **911** (e.g., a GSM transceiver) and/or an unlicensed transceiver **913** (e.g., Wi-Fi, WiMax) for corresponding signal communications. The handset **900** can be a device such as a cellular telephone, a PDA with mobile communications capabilities, and messaging-centric devices. The communications component **910** also facilitates communications reception from terrestrial radio networks (e.g., broadcast), digital satellite radio networks, and Internet-based radio services networks.

[0063] The handset **900** includes a display **912** for displaying text, images, video, telephony functions (e.g., a Caller ID function), setup functions, and for user input. For example, the display **912** can also be referred to as a “screen” that can accommodate the presentation of multimedia content (e.g., music metadata, messages, wallpaper, graphics, etc.). The display **912** can also display videos and can facilitate the generation, editing and sharing of video quotes. A serial I/O interface **914** is provided in communication with the processor **902** to facilitate wired and/or wireless serial communications (e.g., USB, and/or IEEE 1394) through a hardwire connection, and other serial input devices (e.g., a keyboard, keypad, and mouse). This can support updating and troubleshooting the handset **900**, for example. Audio capabilities are provided with an audio I/O component **916**, which can include a speaker for the output of audio signals related to, for example, indication that the user pressed the proper key or key combination to initiate the user feedback signal. The audio I/O component **916** also facilitates the input of audio signals through a microphone to record data and/or telephony voice data, and for inputting voice signals for telephone conversations.

[0064] The handset **900** can include a slot interface **918** for accommodating a SIC (Subscriber Identity Component) in the form factor of a card Subscriber Identity Module (SIM) or universal SIM **920**, and interfacing the SIM card **920** with the processor **902**. However, it is to be appreciated that the SIM card **920** can be manufactured into the handset **900**, and updated by downloading data and software.

[0065] The handset **900** can process IP data traffic through the communications component **910** to accommodate IP traffic from an IP network such as, for example, the Internet, a corporate intranet, a home network, a person area network, etc., through an ISP or broadband cable provider. Thus, VoIP traffic can be utilized by the handset **900** and IP-based multimedia content can be received in either an encoded or decoded format.

[0066] A video processing component **922** (e.g., a camera) can be provided for decoding encoded

multimedia content. The video processing component **922** can aid in facilitating the generation, editing, and sharing of video quotes. The handset **900** also includes a power source **924** in the form of batteries and/or an AC power subsystem, which power source **924** can interface to an external power system or charging equipment (not shown) by a power I/O component **926**.

[0067] The handset **900** can also include a video component **930** for processing video content received and, for recording and transmitting video content. For example, the video component **930** can facilitate the generation, editing and sharing of video quotes. A location tracking component **932** facilitates geographically locating the handset **900**. As described hereinabove, this can occur when the user initiates the feedback signal automatically or manually. A user input component **934** facilitates the user initiating the quality feedback signal. The user input component **934** can also facilitate the generation, editing and sharing of video quotes. The user input component **934** can include such conventional input device technologies such as a keypad, keyboard, mouse, stylus pen, and/or touchscreen, for example.

[0068] Referring again to the applications **906**, a hysteresis component **936** facilitates the analysis and processing of hysteresis data, which is utilized to determine when to associate with the access point. A software trigger component **938** can be provided that facilitates triggering of the hysteresis component **936** when the Wi-Fi transceiver **913** detects the beacon of the access point. A SIP client **940** enables the handset **900** to support SIP protocols and register the subscriber with the SIP registrar server. The applications **906** can also include a client **942** that provides at least the capability of discovery, play and store of multimedia content, for example, music.

[0069] The handset **900**, as indicated above related to the communications component **910**, includes an indoor network radio transceiver **913** (e.g., Wi-Fi transceiver). This function supports the indoor radio link, such as IEEE 802.11, for the dual-mode GSM handset **900**. The handset **900** can accommodate at least satellite radio services through a handset that can combine wireless voice and digital radio chipsets into a single handheld device.

[0070] In order to provide additional context for various embodiments described herein, FIG. **10** and the following discussion are intended to provide a brief, general description of a suitable computing environment **1000** in which the various embodiments of the embodiment described herein can be implemented. While the embodiments have been described above in the general context of computer-executable instructions that can run on one or more computers, those skilled in the art will recognize that the embodiments can be also implemented in combination with other program modules and/or as a combination of hardware and software.

[0071] Generally, program modules include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the disclosed methods can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, minicomputers, mainframe computers, Internet of Things (IoT) devices, distributed computing systems, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0072] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0073] Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media or machine-

readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data or unstructured data.

[0074] Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0075] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0076] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0077] With reference again to FIG. 10, the example environment **1000** for implementing various embodiments of the aspects described herein includes a computer **1002**, the computer **1002** including a processing unit **1004**, a system memory **1006** and a system bus **1008**. The system bus **1008** couples system components including, but not limited to, the system memory **1006** to the processing unit **1004**. The processing unit **1004** can be any of various commercially available processors. Dual microprocessors and other multi-processor architectures can also be employed as the processing unit **1004**.

[0078] The system bus **1008** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **1006** includes ROM **1010** and RAM **1012**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **1002**, such as during startup. The RAM **1012** can also include a high-speed RAM such as static RAM for caching data.

[0079] The computer **1002** further includes an internal hard disk drive (HDD) **1014** (e.g., EIDE, SATA), one or more external storage devices **1016** (e.g., a magnetic floppy disk drive (FDD) **1016**, a memory stick or flash drive reader, a memory card reader, etc.) and an optical disk drive **1020** (e.g., which can read or write from a CD-ROM disc, a DVD, a BD, etc.). While the internal HDD **1014** is illustrated as located within the computer **1002**, the internal HDD **1014** can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment **1000**, a solid state drive (SSD) could be used in addition to, or in place of, an HDD **1014**. The HDD **1014**, external storage device(s) **1016** and optical disk drive **1020** can be connected to the system bus **1008** by an HDD interface **1024**, an external storage interface **1026** and an optical drive interface **1028**, respectively. The interface **1024** for external drive

implementations can include at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0080] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **1002**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0081] A number of program modules can be stored in the drives and RAM **1012**, including an operating system **1030**, one or more application programs **1032**, other program modules **1034** and program data **1036**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **1012**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0082] Computer **1002** can optionally comprise emulation technologies. For example, a hypervisor (not shown) or other intermediary can emulate a hardware environment for operating system **1030**, and the emulated hardware can optionally be different from the hardware illustrated in FIG. **10**. In such an embodiment, operating system **1030** can comprise one virtual machine (VM) of multiple VMs hosted at computer **1002**. Furthermore, operating system **1030** can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications **1032**. Runtime environments are consistent execution environments that allow applications **1032** to run on any operating system that includes the runtime environment. Similarly, operating system **1030** can support containers, and applications **1032** can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

[0083] Further, computer **1002** can be enable with a security module, such as a trusted processing module (TPM). For instance, with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer **1002**, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

[0084] A user can enter commands and information into the computer **1002** through one or more wired/wireless input devices, e.g., a keyboard **1038**, a touch screen **1040**, and a pointing device, such as a mouse **1042**. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit **1004** through an input device interface **1044** that can be coupled to the system bus **1008**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a USB port, an IR interface, a BLUETOOTH® interface, etc.

[0085] A monitor **1046** or other type of display device can be also connected to the system bus **1008** via an interface, such as a video adapter **1048**. In addition to the monitor **1046**, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

[0086] The computer **1002** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **1050**. The remote computer(s) **1050** can be a workstation, a server computer, a router,

a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **1002**, although, for purposes of brevity, only a memory/storage device **1052** is illustrated. The logical connections depicted include wired/wireless connectivity to a local area network (LAN) **1054** and/or larger networks, e.g., a wide area network (WAN) **1056**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0087] When used in a LAN networking environment, the computer **1002** can be connected to the local network **1054** through a wired and/or wireless communication network interface or adapter **1058**. The adapter **1058** can facilitate wired or wireless communication to the LAN **1054**, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter **1058** in a wireless mode.

[0088] When used in a WAN networking environment, the computer **1002** can include a modem **1060** or can be connected to a communications server on the WAN **1056** via other means for establishing communications over the WAN **1056**, such as by way of the Internet. The modem **1060**, which can be internal or external and a wired or wireless device, can be connected to the system bus **1008** via the input device interface **1044**. In a networked environment, program modules depicted relative to the computer **1002** or portions thereof, can be stored in the remote memory/storage device **1052**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0089] When used in either a LAN or WAN networking environment, the computer **1002** can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices **1016** as described above. Generally, a connection between the computer **1002** and a cloud storage system can be established over a LAN **1054** or WAN **1056** e.g., by the adapter **1058** or modem **1060**, respectively. Upon connecting the computer **1002** to an associated cloud storage system, the external storage interface **1026** can, with the aid of the adapter **1058** and/or modem **1060**, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface **1026** can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer **1002**.

[0090] The computer **1002** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0091] Wi-Fi, or Wireless Fidelity, allows connection to the Internet from a couch at home, in a hotel room, or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands, at a 7 Mbps (802.11a) or 54 Mbps (802.11b) data rate, for example, or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 16BaseT wired Ethernet networks used in many offices.

[0092] An aspect of 5G, which differentiates from previous 4G systems, is the use of NR. NR architecture can be designed to support multiple deployment cases for independent configuration of

resources used for RACH procedures. Since the NR can provide additional services than those provided by LTE, efficiencies can be generated by leveraging the pros and cons of LTE and NR to facilitate the interplay between LTE and NR, as discussed herein.

[0093] Reference throughout this specification to “one embodiment,” or “an embodiment,” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase “in one embodiment,” “in one aspect,” or “in an embodiment,” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics can be combined in any suitable manner in one or more embodiments.

[0094] As used in this disclosure, in some embodiments, the terms “component,” “system,” “interface,” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution, and/or firmware. As an example, a component can be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component.

[0095] One or more components can reside within a process and/or thread of execution and a component can be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components can communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software application or firmware application executed by one or more processors, wherein the processor can be internal or external to the apparatus and can execute at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confer(s) at least in part the functionality of the electronic components. In an aspect, a component can emulate an electronic component via a virtual machine, e.g., within a cloud computing system. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0096] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0097] Moreover, terms such as “mobile device equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “communication device,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a

subscriber or mobile device of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings. Likewise, the terms “access point (AP),” “Base Station (BS),” BS transceiver, BS device, cell site, cell site device, “Node B (NB),” “evolved Node B (eNode B),” “home Node B (HNB)” and the like, are utilized interchangeably in the application, and refer to a wireless network component or appliance that transmits and/or receives data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream from one or more subscriber stations. Data and signaling streams can be packetized or frame-based flows.

[0098] Furthermore, the terms “device,” “communication device,” “mobile device,” “subscriber,” “customer entity,” “consumer,” “customer entity,” “entity” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0099] Embodiments described herein can be exploited in substantially any wireless communication technology, comprising, but not limited to, wireless fidelity (Wi-Fi), global system for mobile communications (GSM), universal mobile telecommunications system (UMTS), worldwide interoperability for microwave access (WiMAX), enhanced general packet radio service (enhanced GPRS), third generation partnership project (3GPP) long term evolution (LTE), third generation partnership project 2 (3GPP2) ultra mobile broadband (UMB), high speed packet access (HSPA), Z-Wave, Zigbee and other 802.XX wireless technologies and/or legacy telecommunication technologies.

[0100] The various aspects described herein can relate to New Radio (NR), which can be deployed as a standalone radio access technology or as a non-standalone radio access technology assisted by another radio access technology, such as Long Term Evolution (LTE), for example. It should be noted that although various aspects and embodiments have been described herein in the context of 6G, Universal Mobile Telecommunications System (UMTS), and/or Long Term Evolution (LTE), 5G, or other next generation networks, the disclosed aspects are not limited to 6G, a UMTS implementation, and/or an LTE implementation as the techniques can also be applied in 3G, 4G, 5G, or LTE systems. For example, aspects or features of the disclosed embodiments can be exploited in substantially any wireless communication technology. Such wireless communication technologies can include UMTS, Code Division Multiple Access (CDMA), Wi-Fi, Worldwide Interoperability for Microwave Access (WiMAX), General Packet Radio Service (GPRS), Enhanced GPRS, Third Generation Partnership Project (3GPP), LTE, Third Generation Partnership Project 2 (3GPP2) Ultra Mobile Broadband (UMB), High Speed Packet Access (HSPA), Evolved High Speed Packet Access (HSPA+), High-Speed Downlink Packet Access (HSDPA), High-Speed Uplink Packet Access (HSUPA), Zigbee, or another IEEE 802.XX technology. Additionally, substantially all aspects disclosed herein can be exploited in legacy telecommunication technologies.

[0101] As used herein, the term “infer” or “inference” refers generally to the process of reasoning about, or inferring states of, the system, environment, user, and/or intent from a set of observations as captured via events and/or data. Captured data and events can include user data, device data, environment data, data from sensors, sensor data, application data, implicit data, explicit data, etc. Inference can be employed to identify a specific context or action, or can generate a probability distribution over states of interest based on a consideration of data and events, for example.

[0102] Inference can also refer to techniques employed for composing higher-level events from a set of events and/or data. Such inference results in the construction of new events or actions from a set of observed events and/or stored event data, whether the events are correlated in close temporal proximity, and whether the events and data come from one or several event and data sources.

Various classification procedures and/or systems (e.g., support vector machines, neural networks, expert systems, Bayesian belief networks, fuzzy logic, and data fusion engines) can be employed in connection with performing automatic and/or inferred action in connection with the disclosed subject matter.

[0103] In addition, the various embodiments can be implemented as a method, apparatus, or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device, machine-readable device, computer-readable carrier, computer-readable media, machine-readable media, computer-readable (or machine-readable) storage/communication media. For example, computer-readable media can comprise, but are not limited to, a magnetic storage device, e.g., hard disk; floppy disk; magnetic strip(s); an optical disk (e.g., compact disk (CD), a digital video disc (DVD), a Blu-ray Disc™ (BD)); a smart card; a flash memory device (e.g., card, stick, key drive); and/or a virtual device that emulates a storage device and/or any of the above computer-readable media. Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0104] The above description of illustrated embodiments of the subject disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosed embodiments to the precise forms disclosed. While specific embodiments and examples are described herein for illustrative purposes, various modifications are possible that are considered within the scope of such embodiments and examples, as those skilled in the relevant art can recognize.

[0105] In this regard, while the subject matter has been described herein in connection with various embodiments and corresponding figures, where applicable, it is to be understood that other similar embodiments can be used or modifications and additions can be made to the described embodiments for performing the same, similar, alternative, or substitute function of the disclosed subject matter without deviating therefrom. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims below.

Claims

1. A method, comprising: in response to determining, by a first network device comprising a processor, that a user equipment in communication with the first network device is to be handed off to a second network device: generating, by the first network device, first service application package data representative of first utilization of a first service by the user equipment while in communication with the first network device; generating, by the first network device, second service application package data representative of second utilization of a second service by the user equipment while in communication with the first network device; aggregating, by the first network device, the first service application package data and the second service application package data, resulting in aggregated service application package data; and sending, by the first network device, the aggregated service application package data to the second network device or a first mobile application function (MAF) component associated with the second network device.
2. The method of claim 1, wherein the determining comprises determining that the user equipment is to be handed off to the second network device based on location data associated with the user equipment, where the location data includes a physical movement of the user equipment.
3. The method of claim 1, wherein the determining comprises determining that the user equipment is to be handed off to the second network device based on a predicted route of the user equipment, wherein the first MAF component includes additional data related to a subscriber of the user equipment.

- 4.** The method of claim 1, wherein the first service application package data comprises: first user equipment information obtained during the first utilization of the first service by the user equipment while in communication with the first network device.
- 5.** The method of claim 1, wherein the first service application package data comprises: microservice information specifying a group of microservice applications to be instantiated at the second network device for utilization of the first service by the user equipment while in communication with the second network device.
- 6.** The method of claim 1, further comprising facilitating, by the first network device, a communication handover of the user equipment from the first network device to the second network device.
- 7.** The method of claim 1, further comprising: determining that the user equipment is to be transferred from a first location to a second location within a coverage area of the first network device; and triggering a second MAF component associated with a first location coverage area to send a request to push subscriber data of the user equipment to a third MAF component associated with a second location coverage area.
- 8.** A first mobile edge computing device, comprising: a processor; and a memory that stores executable instructions that, when executed by the processor, facilitate performance of operations, comprising: in response to determining that a mobile device in communication with the first mobile edge computing device is to be handed off to a second mobile edge computing device: generating first service application package data representative of first utilization of a first service by the mobile device while in communication with the first mobile edge computing device; generating second service application package data representative of second utilization of a second service by the mobile device while in communication with the first mobile edge computing device; aggregating the first service application package data and the second service application package data, resulting in aggregated service application package data; and transmitting the aggregated service application package data to the second mobile edge computing device or a first mobile application function (MAF) component associated with the second mobile edge computing device.
- 9.** The first mobile edge computing device of claim 8, wherein the determining comprises determining that the mobile device is to be handed off to the second mobile edge computing device based on global positioning system coordinates associated with the mobile device.
- 10.** The first mobile edge computing device of claim 8, wherein the first service application package data comprises: mobile device information obtained during the first utilization of the first service by the mobile device while in communication with the first mobile edge computing device.
- 11.** The first mobile edge computing device of claim 8, wherein the first service application package data comprises: microservice information specifying a group of microservice applications to be instantiated at the second mobile edge computing device for utilization of the first service by the mobile device while in communication with the second mobile edge computing device.
- 12.** The first mobile edge computing device of claim 8, wherein the second service application package data comprises: mobile device information obtained during the second utilization of the second service by the mobile device while in communication with the first mobile edge computing device.
- 13.** The first mobile edge computing device of claim 8, wherein the second service application package data comprises: microservice information specifying a group of microservice applications to be instantiated at the second mobile edge computing device for utilization of the second service by the mobile device while in communication with the second mobile edge computing device.
- 14.** The first mobile edge computing device of claim 8, wherein the operations further comprise facilitating a handover of the mobile device from the first mobile edge computing device to the second mobile edge computing device.
- 15.** A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processor of a first radio network node, facilitate performance of operations,

comprising: in response to determining that a user equipment in communication with the first radio network node is to be transferred from being serviced via the first radio network node to being serviced via a second radio network node: generating first service application package data representative of first utilization of a first service by the user equipment concurrently with the user equipment being in communication with the first radio network node; generating second service application package data representative of second utilization of a second service by the user equipment concurrently with the user equipment being in communication with the first radio network node; aggregating the first service application package data and the second service application package data, resulting in aggregated service application package data; and transmitting the aggregated service application package data to the second radio network node or a first mobile application function (MAF) component associated with the second radio network node.

16. The non-transitory machine-readable medium of claim 15, wherein the operations further comprise determining that the user equipment is to be transferred from a first location to a second location within a coverage area being serviced via the first radio network node based on movement data associated with the user equipment.

17. The non-transitory machine-readable medium of claim 16, wherein the first service application package data comprises: user equipment information obtained during the first utilization of the first service by the user equipment concurrently with the user equipment being in communication with the first radio network node; wherein the operations further comprise triggering a second MAF component associated with a first location coverage area to send a request to push subscriber data of the user equipment to a third MAF component associated with a second location coverage area.

18. The non-transitory machine-readable medium of claim 17, wherein the first service application package data comprises: microservice information specifying a group of microservice applications to be instantiated at the second radio network node for utilization of the first service by the user equipment concurrently with the user equipment being in communication with the second radio network node; wherein the operations further comprise sending the subscriber data of the user equipment to the third MAF component via a service related application package (SRAP).

19. The non-transitory machine-readable medium of claim 15, wherein the second service application package data comprises: user equipment information obtained during the second utilization of the second service by the user equipment concurrently with the user equipment being in communication with the first radio network node.

20. The non-transitory machine-readable medium of claim 18, wherein the second service application package data comprises: microservice information specifying a group of microservice applications to be instantiated at the second radio network node for utilization of the second service by the user equipment concurrently with the user equipment being in communication with the second radio network node; and wherein the operations further comprise: requesting an acknowledgement from the third MAF component that has accepted a data transfer; and upon receiving the acknowledgement, sending the subscriber data of the user equipment to the third MAF component via a service related application package (SRAP).
